A 6/23 BOUND FOR PENNY GRAPHS

TOM DE GROOT

Abstract. A Euclidean minimum-distance graph joins pairs at distance one in a finite point set in the plane whose mutual distances are at least one. We prove that every such graph on n vertices has an independent set of size at least $\lceil 6n/23 \rceil$, improving the author's earlier $\lceil 7n/27 \rceil$ bound. The proof is self-contained; its new ingredient is the use of equality in a local expansion estimate to force a finite patch of the triangular lattice.

1. Introduction

Let $P \subset \mathbb{R}^2$ be finite and suppose that $|x - y| \geq 1$ whenever $x, y \in P$ are distinct. Its *minimum-distance graph* $G(P)$ has vertex set P, with two points adjacent precisely when their distance is one. Equivalently, these are the contact graphs of packings of congruent closed discs of radius $1/2$. Write

$$g(n) = \min\{\alpha(G(P)) : |P| = n\},$$

where α denotes the independence number. Erdős asked for estimates of this function [3]. Planarity gives $g(n) \geq n/4$ [5]; Csizmadia improved this to $9n/35$ [1], and Swanepoel obtained $8n/31$ [6]. In an earlier preprint, the present author proved $g(n) \geq \lceil 7n/27 \rceil$ [2]. The construction of Pach and Tóth gives $g(n) \leq 5n/16 + O(1)$ [4].

Theorem 1.1. *Every Euclidean minimum-distance graph G satisfies*

$$\alpha(G) \geq \left\lceil \frac{6|V(G)|}{23} \right\rceil.$$

Consequently, $g(n) \geq \lceil 6n/23 \rceil$ for every n.

The proof proceeds in four stages. First, a smallest counterexample must expand every independent set of at most six vertices by a factor of at least three. Second, a boundary-curvature count produces nine consecutive degree-four boundary vertices, each incident with two boundary triangles, whose total clockwise turn is less than $\pi/3$. Third, Euclidean angle estimates show that the next layer of this strip has a unique break, and that the break can occur only in one of its first three positions. Finally, explicit independent sets force a flat substrip; equality in the six-vertex expansion bound then forces successive triangular-lattice vertices until a last independent set contradicts minimality.

The improvement over $7/27$ does not come from extending the earlier case analysis. The numerical gain is $6/23 - 7/27 = 1/621$, and the associated deletion threshold changes from $7|N(S)| \leq 20|S|$ to the stricter inequality $6|N(S)| \leq 17|S|$. The earlier proof uses expansion for independent sets of size at most seven, obtains an eleven-vertex boundary chain, locates the unique break at a fixed position, and analyzes the occupancy patterns of three sectors in a longer flat strip. Here expansion is available only for sets of size at most six, so the boundary chain has nine vertices and the break is initially confined only to the first three positions. The decisive new step is to exploit equality: when a six-vertex independent set has an a priori neighbourhood bound of eighteen, expansion forces all eighteen neighbours to occur. The remaining available unit-neighbour directions are then occupied, and iterating this observation creates the finite lattice patch used in the final contradiction.

2020 *Mathematics Subject Classification.* 05C69, 52C10, 52C15.

Key words and phrases. penny graph, minimum-distance graph, independent set, contact graph.

The boundary reduction is adapted from the method of Csizmadia and Swanepoel, but every statement used below is proved in the Euclidean setting. All graphs are finite and simple. For $S \subseteq V(G)$, let

$$N_G(S) = \{v \in V(G) \setminus S : v \text{ has a neighbour in } S\},$$

and omit the subscript when the graph is clear.

2. Minimality and boundary curvature

Assume for a contradiction that Theorem 1.1 is false, and choose a counterexample G with the fewest vertices. Put $n = |V(G)|$.

Lemma 2.1. *If $\varnothing \neq S \subseteq V(G)$ is independent and $6|N(S)| \leq 17|S|$, then G is not a counterexample.*

Proof. Let $G' = G - (S \cup N(S))$. By minimality, $\alpha(G') \geq 6|V(G')|/23$, and no vertex of G' is adjacent to S. Hence

$$\alpha(G) \geq |S| + \alpha(G') \geq \frac{6n + 17|S| - 6|N(S)|}{23} \geq \frac{6n}{23},$$

a contradiction. □

Corollary 2.2. *Every independent set S with $1 \leq |S| \leq 6$ satisfies $|N(S)| \geq 3|S|$. In particular, G has minimum degree at least three.*

Proof. If $|S| = k \leq 6$ and $|N(S)| \leq 3k - 1$, then $6|N(S)| \leq 18k - 6 \leq 17k$, contrary to Lemma 2.1. The degree assertion is the case $k = 1$. □

Lemma 2.3. *The straight-line drawing of G is planar, every vertex has degree at most six, and G is two-connected. Consequently, the boundary of the outer face is a simple cycle.*

Proof. Suppose that two unit segments cross properly at z. Choose an endpoint x of the first and an endpoint y of the second with $|x - z|, |y - z| \leq 1/2$. Then $|x - y| \leq 1$, and equality in the triangle inequality would make the two segments collinear. Thus $|x - y| < 1$, which is impossible.

The neighbours of any vertex lie on its unit circle. The smaller angular separation between two of them is at least $\pi/3$, since their mutual distance is at least one. Hence every vertex has degree at most six.

If G were disconnected, independent sets supplied by minimality in its components could be united. Suppose instead that x is a cut vertex. Partition the components of $G - x$ into two nonempty unions with vertex sets A, B, and put $H_A = G[A \cup \{x\}]$ and $H_B = G[B \cup \{x\}]$. Since adjoining one vertex changes the independence number by at most one, either $\alpha(H_A) = \alpha(G[A])$ or $\alpha(H_A) = \alpha(G[A]) + 1$.

In the first case, a maximum independent set of H_A can be chosen to avoid x, and hence can be united with one in $G[B]$. Minimality gives

$$\alpha(G) \geq \alpha(H_A) + \alpha(G[B]) \geq \frac{6(|A| + 1) + 6|B|}{23} = \frac{6n}{23}.$$

In the second case, choose a maximum independent set of H_A containing x. After removing x, it is a maximum independent set of $G[A]$ and contains no neighbour of x, so it can be united with any independent set of H_B. Thus

$$\alpha(G) \geq \alpha(G[A]) + \alpha(H_B) \geq \frac{6|A| + 6(|B| + 1)}{23} = \frac{6n}{23}.$$

Both alternatives contradict the choice of G. A two-connected plane graph has a simple outer-face boundary cycle. □

We shall use two elementary angle facts. All angles are ordinary Euclidean angles in $[0, \pi]$.

Lemma 2.4. *Let a, b, c, d occur in this order on a simple quadrilateral and suppose*

$$|a - b| = |b - c| = |c - d| = 1, \qquad |a - c|, |a - d|, |b - d| \geq 1.$$

Then $\angle abc + \angle bcd \geq \pi$.

Proof. Put $b = (0, 0)$, $c = (1, 0)$, and write $\beta = \angle abc$, $\gamma = \angle bcd$. The inequalities $|a-c|, |b-d| \geq 1$ give $\beta, \gamma \geq \pi/3$. After reflection in the line bc, take $a = (\cos\beta, \sin\beta)$.

If a, d lie on the same side of bc, then $d = (1 - \cos\gamma, \sin\gamma)$. Were $\beta + \gamma < \pi$, then $|a - d|^2 = 3 + 2\cos(\beta + \gamma) - 2\cos\beta - 2\cos\gamma < 1$, because $\cos\beta + \cos\gamma > 1 + \cos(\beta + \gamma)$. If a, d lie on opposite sides, take $d = (1 - \cos\gamma, -\sin\gamma)$. Under the same assumption, the segment ad meets the line bc at abscissa

$$\xi = \frac{\sin\gamma\cos\beta + \sin\beta(1 - \cos\gamma)}{\sin\beta + \sin\gamma}.$$

The numerator and the difference between denominator and numerator are

$$2\sin(\gamma/2)\cos(\beta - \gamma/2), \qquad 2\sin(\beta/2)\cos(\gamma - \beta/2),$$

respectively. Both are positive when $\beta, \gamma \geq \pi/3$ and $\beta + \gamma < \pi$. Thus $0 < \xi < 1$, so ad crosses the open segment bc, contrary to simplicity. □

Lemma 2.5. *Let Q be a simple cycle in G, and let H be the subgraph induced by the vertices on or inside Q. If D_j is the number of vertices of degree j in H on Q, then*

$$\sum_{j \geq 2} (4 - j) D_j \geq 6.$$

In particular, $2D_2 + D_3 \geq 6$.

Proof. At a boundary vertex of degree j in H, the interior angle of Q contains $j - 1$ successive gaps between unit-neighbour directions, each at least $\pi/3$. If Q has t vertices, then $(t - 2)\pi \geq (\pi/3)\sum_{j\geq 2}(j - 1)D_j$. Using $t = \sum_j D_j$ gives the first assertion; discarding the nonpositive terms with $j \geq 4$ gives the second. □

We now turn the curvature inequality into a long boundary chain of small total turn. Let $C = v_0 v_1 \cdots v_{t-1} v_0$ be the outer boundary, oriented counterclockwise. A boundary edge is *triangular* if its endpoints have a common neighbour. Such a neighbour is unique and lies on the interior side of the edge: the two endpoints and a common neighbour form a unit equilateral triangle, and an apex on the outer-face side would bound an empty triangular region there.

Write θ_i for the interior angle at v_i and put $\tau_i = \theta_i - \pi$. At a degree-four boundary vertex, $\tau_i \geq 0$. A *boundary defect* is a nontriangular boundary edge or a boundary vertex of degree five or six.

An *anchored triangular run* is a sequence $v_{a+1}, \ldots, v_{a+k}$ of consecutive degree-four boundary vertices such that every edge $v_j v_{j+1}$, $a \leq j \leq a + k$, is triangular, and such that at least one end is anchored as follows. At the left end, $\deg v_a = 3$ and either $\deg v_{a-1} = 3$, or $\deg v_{a-1} = 4$ and $v_{a-1}v_a$ is triangular; the right-hand condition is the reflection of this one. Its length is k, and it is *concave* if $\sum_{j=a+1}^{a+k} \tau_j \geq \pi/3$. Including both boundary edges adjacent to the run ensures that no vertex of the run is incident with a nontriangular boundary edge.

Lemma 2.6. *Let b be the number of nontriangular boundary edges and let d_j be the number of boundary vertices of degree j. Let $\mathcal{R}$ be any family of pairwise vertex-disjoint anchored triangular runs, and let B be the number of concave members of $\mathcal{R}$. Then*

$$d_3 \geq d_5 + 2d_6 + b + B + 6.$$

Proof. At a boundary vertex v of degree j, the interior angle is the sum of $j - 1$ consecutive neighbour-direction gaps. Set $E(v) = \theta(v) - (j - 1)\pi/3 \geq 0$.

For a nontriangular boundary edge uv, let ux and vy be the edges immediately next to uv on the boundary of its bounded incident face. The points x, u, v, y form a simple quadrilateral, up to a collinear limiting case, and Lemma 2.4 gives $\angle xuv + \angle uvy \geq \pi$. Thus these two particular gaps contribute at least $\pi/3$ beyond their baseline $2\pi/3$. Distinct nontriangular boundary edges use distinct neighbour gaps, even when they share an endpoint.

Every concave run contributes at least another $\pi/3$, since $E(v) = \tau(v)$ at a degree-four boundary vertex. The runs are vertex-disjoint, and no run vertex is incident with a nontriangular boundary edge, so these contributions are disjoint. Consequently,

$$(t-2)\pi = \sum_i \theta_i \geq \frac{\pi}{3}\big(2d_3 + 3d_4 + 4d_5 + 5d_6 + b + B\big).$$

Substituting $t = d_3 + d_4 + d_5 + d_6$ and rearranging proves the assertion. ☐

Lemma 2.7. *Let $u = x_0, x_1, \ldots, x_D = v$ be the forward boundary path between two consecutive degree-three vertices, and let x_{D+1} be the next boundary vertex after v. Suppose that this interval contains no anchored triangular run of length at least nine and no boundary defect, except possibly the edge x_0x_1 or the vertex x_1. Then at least one of x_{D+1} and vx_{D+1} is a boundary defect, unless $D = 1$ and uv is nontriangular.*

Proof. Assume that neither x_{D+1} nor vx_{D+1} is a defect. Then $\deg x_{D+1} \leq 4$, the edge vx_{D+1} is triangular, every vertex $x_2, \ldots, x_{D-1}$ has degree four, and every boundary edge from x_1x_2 through vx_{D+1} is triangular. If $D \geq 3$, the sequence $x_2, \ldots, x_{D-1}$ is a right-anchored triangular run of length $D-2$. Hence $D \leq 10$.

Put $k = \lceil D/2 \rceil + 1 \leq 6$ and take

$$S = \{x_0, x_2, x_4, \ldots, x_{2\lceil D/2 \rceil}\}.$$

Independence. If D is even, the last member is v; if D is odd, it is x_{D+1}. In the odd case, if $D = 1$ and uv is nontriangular, this is the stated exception. Otherwise the last two selected vertices share both v and the third neighbour of v: the two boundary edges at v are triangular and $\deg v = 3$. They are therefore the two equilateral apices on opposite sides of the unit segment joining these common neighbours, and their distance is $\sqrt{3}$.

No other selected pair is adjacent. Indeed, such an edge, together with the intervening outer-boundary path, would form a simple cycle Q. Let H be induced by the vertices on or inside Q. Every internal vertex of the boundary subpath retains all its incident edges in H. A chord endpoint has degree at least two in H, and at least three unless it is one of u, v, x_{D+1}; the third edge is supplied by the equilateral triangle on the adjacent triangular boundary edge. Every other vertex of Q has degree at least four in H, apart from an internal occurrence of one of the three named vertices, which has degree three. Hence $\sum_{z \in V(Q)}(4 - \deg_H z) \leq 5$, contrary to Lemma 2.5. Applied to the outer boundary, the same lemma shows that there are at least six degree-three boundary vertices. Hence an interval between consecutive degree-three vertices, together with its successor, cannot wrap around and repeat a selected vertex. Thus S is independent.

Neighbourhood count. If D is even, the selected degrees sum to at most $4k-2$, and consecutive selected vertices share the skipped boundary vertex, so $|N(S)| \leq (4k-2) - (k-1) = 3k-1$. If D is odd and is not exceptional, the degree sum is at most $4k-1$; the last pair has two common neighbours and every earlier consecutive pair has one, so again $|N(S)| \leq 3k-1$. Both bounds contradict Corollary 2.2. ☐

Lemma 2.8. *There are consecutive boundary vertices $p_0, p_1, \ldots, p_9$ such that*

$$\deg p_0 = 3, \qquad \deg p_i = 4 \ (1 \leq i \leq 9), \qquad \sum_{i=1}^{9} \tau_i < \frac{\pi}{3},$$

and every edge p_ip_{i+1}, $0 \leq i \leq 8$, is triangular. If p_{-1} is the predecessor of p_0, then either $\deg p_{-1} = 3$, or $\deg p_{-1} = 4$ and $p_{-1}p_0$ is triangular.

Proof. List the degree-three boundary vertices cyclically as $u_1, \ldots, u_s$, with $u_{s+1} = u_1$. They divide the boundary into $s = d_3$ forward intervals. Assign each defective vertex to the interval in whose interior it lies and each defective edge to the interval containing it. Let $N = b + d_5 + d_6$ be the number of defects. In every interval containing an anchored triangular run of length at least nine, choose one such run. Let A be the number chosen and B the number of chosen runs that are concave. The chosen runs are pairwise vertex-disjoint.

Claim. $N + A \geq d_3$. If every interval contains a defect or a chosen run, there is nothing to prove. Otherwise relabel so that the first interval, from u_1 to u_2, contains neither. For $1 \leq j \leq s$, let M_j count the defects and chosen runs in the first j intervals. We prove inductively that

$M_j \geq j-1$, with the following strengthening: if equality holds, then the first edge or first internal vertex of the next interval is defective.

For $j=1$, this is Lemma 2.7; its exceptional case cannot occur because the first edge is not defective. Suppose the strengthened assertion holds for $j-1$. If $M_{j-1} \geq j-1$, then $M_j \geq j-1$. Equality at stage j can occur only when the j-th interval contains neither a defect nor a chosen run, and Lemma 2.7 then places a defect at the beginning of the next interval. If instead $M_{j-1} = j-2$, the induction hypothesis already places a defect at the beginning of the j-th interval. Hence $M_j \geq j-1$. If equality holds, this is the only defect in that interval and the interval contains no chosen run; Lemma 2.7 again places a defect at the beginning of the next interval, except possibly when the current interval is a single nontriangular edge joining its degree-three endpoints.

We exclude that last possibility by combining it with the preceding interval. Since $M_{j-1} = j-2$ and $M_{j-2} \geq j-3$, the preceding interval contains no chosen run and at most one defect; if a defect occurs, the induction hypothesis places it at the beginning. Write this interval as $y_0, y_1, \ldots, y_E = u_j$, where y_0 and u_j are consecutive degree-three vertices, and suppose that the current interval is the nontriangular edge $u_j u_{j+1}$. The vertices $y_2, \ldots, y_{E-1}$, when present, form a right-anchored triangular run, so $E \leq 10$. If E is even, take $S = \{y_0, y_2, \ldots, y_E\}$; if E is odd, take $S = \{y_0, y_2, \ldots, y_{E-1}, u_{j+1}\}$. Then $|S| \leq 6$. A chord between selected vertices and the intervening boundary path would again give a cycle whose boundary deficit is at most five, contradicting Lemma 2.5; in the odd case, an edge from the last selected boundary vertex to u_{j+1} would make $u_j u_{j+1}$ triangular. Thus S is independent. Its two endpoints have degree three, every other selected vertex has degree four, and consecutive selected vertices share the skipped boundary vertex. Hence $|N(S)| \leq 3|S| - 1$, contrary to Corollary 2.2.

The induction is complete. At $j = s$, equality would force a defect at the beginning of the first interval, contrary to its choice. Thus $N + A \geq d_3$. Applying Lemma 2.6 to the chosen runs gives

$$d_3 \geq d_5 + 2d_6 + b + B + 6 = N + d_6 + B + 6.$$

It follows that $A \geq B + 6$, so one chosen run is nonconcave.

Reverse the boundary orientation if necessary so that this run is anchored on the left. Take its first nine vertices as $p_1, \ldots, p_9$, and let p_0 be the degree-three anchor. The predecessor condition is the left anchoring condition. Every turn on the run is nonnegative, so the first nine turns have sum less than $\pi/3$. $\square$

For $0 \leq i \leq 8$, let q_i be the unique common neighbour of p_i, p_{i+1}.

3. The second layer

Fix the boundary chain supplied by Lemma 2.8. We first establish the separation estimates used throughout the rest of the proof, then determine the outward neighbours of the vertices q_i, and finally show that their common-neighbour pattern has a unique break.

Identify $\mathbb{R}^2$ with $\mathbb{C}$, put $\omega = e^{i\pi/3}$, and orient the boundary so that its interior lies to the left. Each q_i is the left equilateral apex of the directed edge $p_i p_{i+1}$. With $e_i = p_{i+1} - p_i$, we therefore have

$$q_i = p_i + \omega e_i, \qquad e_i = e_{i-1} e^{-i\tau_i} \quad (1 \leq i \leq 8). \tag{3.1}$$

All τ_i are nonnegative, and every partial sum of turns in the displayed strip is less than $\pi/3$.

Lemma 3.1. *Normalize $p_0 = 0$ and $e_0 = 1$. Then $\operatorname{Re} p_{-1} < 1/2$. Consequently, p_{-1} is at distance greater than one from every p_j, q_j, and every point t_j defined below, with $j \geq 1$.*

Proof. Since $|p_{-1}| = |p_1| = 1$ and $|p_{-1} - p_1| \geq 1$, we have $\operatorname{Re} p_{-1} \leq 1/2$. Equality would place p_{-1} at an equilateral apex of $p_0 p_1$. The interior apex is q_0 and cannot be the boundary predecessor, while the exterior apex would form an empty triangular face on the outer-face side of $p_0 p_1$. Thus the inequality is strict.

Every later edge direction lies in the clockwise cone of angle less than $\pi/3$ from e_0. Hence $\operatorname{Re} p_j > 1 + (j-1)/2$ for $j \geq 2$, while $\operatorname{Re} q_j \geq \operatorname{Re} p_j + 1/2$ and, from the formula below, $\operatorname{Re} t_j \geq \operatorname{Re} p_j + 1$. The case p_1 follows from $|p_{-1} - p_1| > 1$. $\square$

Lemma 3.2. *For indices in the displayed strip, the following hold.*

(1) *If $b - a \geq 2$, then $|p_b - p_a| > 1$.*
(2) *If $a \notin \{b, b+1\}$, then $|p_a - q_b| > 1$.*
(3) *If $t_b = p_b + \omega(e_b + e_{b+1})$, then $|p_a - t_b| > 1$ for every boundary vertex p_a in the strip.*
(4) *If $b - a \geq 2$, then $|q_b - q_a| > 1$.*
(5) *If t_{a+1} is defined, then $|t_{a+1} - q_a| > 1$.*

Proof. Rotate so that the earliest edge in each calculation has direction one. Every later edge has the form $e^{-i\varphi}$ with $0 \leq \varphi < \pi/3$, so its projection on the initial direction is greater than $1/2$, while the projection of $\omega e^{-i\varphi}$ is at least $1/2$.

For (1), the sum of two consecutive edge vectors has length $2\cos(\tau/2) > \sqrt{3}$; with more terms, projection on the first edge exceeds one. For (2), if $a < b$, project $q_b - p_a = e_a + \cdots + e_{b-1} + \omega e_b$ on e_a. If $a > b+1$, use $p_a - q_b = \overline{\omega} e_b + e_{b+1} + \cdots + e_{a-1}$; the two-term case has length at least $\sqrt{3}$, and longer sums have projection greater than one.

For (3), projection settles the cases $a < b$ and $a > b+2$, while $|p_b - t_b| = |e_b + e_{b+1}| > \sqrt{3}$. If $a = b+1$, then $|p_{b+1} - t_b|^2 = 2 - 2\cos(2\pi/3 - \tau_{b+1}) > 1$; if $a = b+2$, then $p_{b+2} - t_b = \overline{\omega}(e_b + e_{b+1})$. Finally,

$$q_b - q_a = \overline{\omega} e_a + e_{a+1} + \cdots + e_{b-1} + \omega e_b, \qquad t_{a+1} - q_a = \overline{\omega} e_a + \omega e_{a+1} + \omega e_{a+2},$$

and the stated projection bounds prove (4) and (5). □

Lemma 3.3. *For every $1 \leq i \leq 7$, the vertex q_i has exactly two neighbours outside the four reference vertices p_i, p_{i+1}, q_{i-1}, and q_{i+1}. The last two are retained as reference directions even when the corresponding edge is absent. Denote the two further neighbours by r_i, s_i so that the cyclic order around q_i is*

$$s_i, r_i, q_{i-1}, p_i, p_{i+1}, q_{i+1}.$$

Proof. At most two. Suppose first that q_i has three further neighbours a_1, a_2, a_3, in this order through the outward sector from q_{i+1} to q_{i-1}. Put $\rho = \angle a_3 q_i p_i$ and $\sigma = \angle p_{i+1} q_i a_1$. The gaps $a_1 q_i a_2$, $a_2 q_i a_3$, and $p_i q_i p_{i+1}$ are each at least $\pi/3$, so $\rho + \sigma \leq \pi$. The cyclic order and planarity make the quadrilaterals determined by the unit chains q_{i-1}, p_i, q_i, a_3 and $a_1, q_i, p_{i+1}, q_{i+1}$ simple. Lemma 2.4 gives

$$\angle q_i p_i q_{i-1} + \rho \geq \pi, \qquad \angle q_{i+1} p_{i+1} q_i + \sigma \geq \pi.$$

Since $\tau_i + \tau_{i+1} = \angle q_i p_i q_{i-1} + \angle q_{i+1} p_{i+1} q_i - 2\pi/3$, this would imply $\tau_i + \tau_{i+1} \geq \pi/3$, a contradiction. Thus there are at most two further neighbours.

At least two. Suppose that there is at most one. Let

$$B_i = \{p_{i+2}, p_{i-1}, p_{i-3}, \ldots\},$$

where the descending sequence ends at p_0 for odd i and at p_{-1} for even i, and put $S_i = B_i \cup \{q_i\}$. Then $|S_i| = \lfloor i/2 \rfloor + 3 \leq 6$. Lemmas 3.1 and 3.2 show that S_i is independent.

We have $|N(B_i)| \leq 3|B_i| + 1$. If the descending sequence ends at a degree-three vertex, the selected degrees sum to $4|B_i| - 1$, and its $|B_i| - 2$ consecutive pairs share their skipped boundary vertices. If i is even and p_{-1} has degree four, the degree sum is $4|B_i|$, but p_{-1} and p_1 share both p_0 and the third neighbour of the two triangular edges at p_0; this supplies one additional repeated incidence and gives the same bound. The four possible local neighbours $p_i, p_{i+1}, q_{i-1}, q_{i+1}$ of q_i already belong to $N(B_i)$. Under the assumption, q_i adds at most one vertex, and therefore $|N(S_i)| \leq 3|B_i| + 2 = 3|S_i| - 1$, contrary to Corollary 2.2. □

Call i a *break* if $s_i \neq r_{i+1}$. When i is not a break, write $t_i = s_i = r_{i+1}$.

Lemma 3.4. *If t_i is defined, then*

$$t_i = p_i + \omega(e_i + e_{i+1}).$$

Moreover,

$$q_i q_{i+1} \in E(G) \iff \tau_{i+1} = 0, \qquad t_i t_{i+1} \in E(G) \iff \tau_{i+1} + \tau_{i+2} = 0,$$

whenever the second expression is defined. Finally, $|q_{i+2} - t_i| \geq \sqrt{3}$.

Proof. The two intersections of the unit circles centred at q_i, q_{i+1} are p_{i+1} and its reflection in the midpoint of q_iq_{i+1}. Thus $t_i = q_i + q_{i+1} - p_{i+1} = p_i + \omega(e_i + e_{i+1})$. Also

$$q_{i+1} - q_i = \overline{\omega}e_i + \omega e_{i+1}, \qquad t_{i+1} - t_i = \overline{\omega}e_i + \omega e_{i+2}.$$

The angles between the two unit summands are respectively $2\pi/3 - \tau_{i+1}$ and $2\pi/3 - (\tau_{i+1} + \tau_{i+2})$. Their sums have length one exactly when those angles equal $2\pi/3$.

For the final assertion, rotate so that $e_i = 1$, and write $e_{i+1} = e^{-ia}$, $e_{i+2} = e^{-i(a+b)}$, where $a, b \geq 0$ and $a + b < \pi/3$. Then

$$|q_{i+2} - t_i|^2 = 3 + 2H(a, b), \quad H(a, b) = \cos a + \cos(2\pi/3 - b) + \cos(2\pi/3 - a - b).$$

For fixed a, $\partial H/\partial b \geq 0$, so $H(a, b) \geq H(a, 0)$; moreover, $2H(a, 0) = \cos a + \sqrt{3}\sin a - 1 = 2\cos(a - \pi/3) - 1 \geq 0$. □

Three consecutive coincidences leave little angular room around the middle vertex.

Lemma 3.5. *If t_a, t_{a+1}, t_{a+2} are defined, then*

$$\big|N(t_{a+1}) \setminus \{t_a, t_{a+2}, q_{a+1}, q_{a+2}\}\big| \leq 2.$$

Proof. Put $t = t_{a+1}$, $q = q_{a+1}$, and $q' = q_{a+2}$. The strip formula gives $q - t = -\omega e_{a+2}$ and $q' - t = \overline{\omega}e_{a+1}$. Hence the smaller angle $\varphi = \angle qtq'$ equals $\pi/3 + \tau_{a+2} < 2\pi/3$, so no further unit neighbour can lie in that sector.

If three further neighbours lie in the complementary sector, let x be the first after q' and z the last before q. Put $\lambda = \angle xtq'$ and $\rho = \angle qtz$. The two gaps between the three further neighbours have total at least $2\pi/3$, so $\lambda + \rho \leq 4\pi/3 - \varphi$. The local cyclic order makes the quadrilaterals t_a, q, t, z and x, t, q', t_{a+2} simple. Lemma 2.4 gives

$$\angle t_aqt + \rho \geq \pi, \qquad \lambda + \angle tq't_{a+2} \geq \pi.$$

The strip directions give $\angle t_aqt = \pi/3 + \tau_{a+1} + \tau_{a+2}$ and $\angle tq't_{a+2} = \pi/3 + \tau_{a+2} + \tau_{a+3}$. Combining the inequalities yields $\tau_{a+1} + \tau_{a+2} + \tau_{a+3} \geq \pi/3$, contrary to the turn bound. □

The preceding restriction already rules out three coincidences at the left end of the strip.

Lemma 3.6. *The three points t_1, t_2, t_3 cannot all be defined.*

Proof. Assume they are defined. By Lemma 3.5, the set $C = N(t_2) \setminus \{t_1, t_3, q_2, q_3\}$ has size at most two. The set

$$S = \{p_6, q_4, p_3, t_2, q_1, p_0\}$$

is independent: boundary–auxiliary pairs are covered by Lemma 3.2, while q_1q_4, q_1t_2, and q_4t_2 are separated by parts (4), (5), and the final assertion of Lemma 3.4. Its neighbourhood is contained in

$$\{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_1, s_4, t_1, t_3\} \cup C,$$

which has at most seventeen vertices. This contradicts Corollary 2.2. □

We next prove the Euclidean estimate used at a break in the second layer.

Lemma 3.7. *Let A, B, C, D be distinct points such that every mutual distance is at least one,*

$$|A - B| = |C - D| = 1, \qquad |A - D| = 2\sin(\alpha/2), \qquad \frac{\pi}{3} \leq \alpha < \frac{2\pi}{3}.$$

Suppose that B, C lie in the same open half-plane bounded by the line AD, and that the segments AB and CD do not cross. Then A, B, C, D occur in this order on a convex quadrilateral and

$$\angle ABC + \angle BCD \leq \alpha + \frac{2\pi}{3}.$$

Proof. The polygon $ABCD$ is simple: its nonadjacent edges AB, CD do not cross, and the segment BC, lying in one open half-plane bounded by AD, does not meet AD. The vertices A, D are convex. Suppose that B were reflex. In a simple quadrilateral, the diagonal BD then lies inside the quadrilateral and splits the interior angle at B, so

$$\angle ABD + \angle DBC > \pi.$$

Put $r = |B - C|$ and $s = |B - D|$. In the triangle BCD,

$$\cos\angle CBD = \frac{r^2 + s^2 - 1}{2rs} \geq \frac{1}{2},$$

since $r^2 - rs + s^2 = (r - s)^2 + rs \geq 1$. In the triangle ABD, writing $d = |A - D| < \sqrt{3}$, we have $\cos\angle ABD = (1 + s^2 - d^2)/(2s) > -1/2$. Hence $\angle ABD + \angle DBC < \pi$, a contradiction. The same argument excludes a reflex vertex at C; hence the quadrilateral is convex.

Place $A = (0, 0)$, $D = (d, 0)$, and write $x = \angle DAB$, $y = \angle CDA$. Then

$$B = (\cos x, \sin x), \qquad C = (d - \cos y, \sin y), \qquad d = 2\sin(\alpha/2).$$

The inequalities $|B - D|, |A - C| \geq 1$ give $x, y \geq L$, where $L = \arccos(d/2) = \pi/2 - \alpha/2$. If $x + y < 4\pi/3 - \alpha = 2L + \pi/3$, put $u = (x + y)/2$ and $v = (x - y)/2$. Then $L \leq u < L + \pi/6 \leq \pi/2$ and $|v| \leq u - L$. A direct calculation gives

$$|B - C|^2 = d^2 + 4\cos^2 u - 4d\cos u\cos v.$$

For fixed u, the right-hand side is largest when $|v| = u - L$. At such an endpoint one of x, y equals L; suppose $y = L$. Since $d = 2\cos L$, both B, C lie on the unit circle about A, at angular separation $2(u - L) < \pi/3$. Hence $|B - C| < 1$, a contradiction. Therefore $x + y \geq 4\pi/3 - \alpha$, and the angle sum of the quadrilateral gives the result. $\square$

Figure 1 records the local order used in the next lemma.

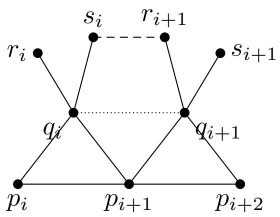

FIGURE 1. The local configuration at a break. The dotted segment joins q_i to q_{i+1}; it need not be an edge. The dashed segment $s_i r_{i+1}$ is not assumed to be an edge.

Lemma 3.8. *If $s_i \neq r_{i+1}$, then*

$$\tau_i + 2\tau_{i+1} + \tau_{i+2} \geq \frac{\pi}{3}, \tag{3.2}$$

and

$$\tau_i + \tau_{i+1} \geq \angle q_{i+1}p_{i+1}r_{i+1}. \tag{3.3}$$

Proof. Use the notation of Figure 1, and put

$$\alpha = \angle q_{i+1}p_{i+1}q_i, \quad \alpha' = \angle q_i p_i q_{i-1}, \quad \alpha'' = \angle q_{i+2}p_{i+2}q_{i+1},$$
$$\beta = \angle p_{i+1}q_i s_i, \quad \beta' = \angle r_i q_i p_i, \quad \gamma = \angle q_i s_i r_{i+1},$$
$$\delta = \angle s_i r_{i+1} q_{i+1}, \quad \varepsilon = \angle r_{i+1} q_{i+1} p_{i+1}, \quad \varepsilon' = \angle p_{i+2} q_{i+1} s_{i+1}.$$

Lemma 2.4 gives $\alpha' + \beta' \geq \pi$ and $\alpha'' + \varepsilon' \geq \pi$. The gaps between the two outward neighbours of q_i and of q_{i+1} are at least $\pi/3$, so $\beta' \leq 4\pi/3 - \beta$ and $\varepsilon' \leq 4\pi/3 - \varepsilon$.

The cyclic orders place s_i, r_{i+1} in the same half-plane bounded by $q_i q_{i+1}$, opposite p_{i+1}, and the unit edges $q_i s_i, q_{i+1} r_{i+1}$ do not cross. Here $\alpha = \pi/3 + \tau_{i+1} < 2\pi/3$ and the isosceles triangle $q_i p_{i+1} q_{i+1}$ gives $|q_i - q_{i+1}| = 2\sin(\alpha/2)$. Lemma 3.7, applied to $(A, B, C, D) = (q_i, s_i, r_{i+1}, q_{i+1})$, shows that $q_i, s_i, r_{i+1}, q_{i+1}$ form a convex quadrilateral and that $\gamma + \delta \leq \alpha + 2\pi/3$. The pentagon $p_{i+1}q_{i+1}r_{i+1}s_i q_i$ is simple, so $\alpha + \beta + \gamma + \delta + \varepsilon = 3\pi$. Since $\theta_i = \alpha' + 2\pi/3$, $\theta_{i+1} = \alpha + 2\pi/3$, and $\theta_{i+2} = \alpha'' + 2\pi/3$, we obtain in one chain

$$\tau_i + 2\tau_{i+1} + \tau_{i+2} \geq 2\alpha - \beta' - \varepsilon' + \frac{2\pi}{3} \geq \alpha - \gamma - \delta + \pi \geq \frac{\pi}{3},$$

which proves (3.2).

For (3.3), put

$$a = \angle s_i p_{i+1} q_i, \quad h = \angle r_{i+1} p_{i+1} s_i, \quad b = \angle q_{i+1} p_{i+1} r_{i+1}, \quad d = |p_{i+1} - r_{i+1}|.$$

Then $0 \le a \le \pi/3$, $1 \le d \le 2$, and

$$\tau_{i+1} = a + h + b - \frac{\pi}{3}. \tag{3.4}$$

Let $\rho = \angle s_i q_i r_i \ge \pi/3$. Around q_i, $\beta' + \pi/3 + (\pi - 2a) + \rho = 2\pi$, so $\beta' \le \pi/3 + 2a$. Lemma 2.4 at q_{i-1}, p_i, q_i, r_i gives $\alpha' + \beta' \ge \pi$; since $\tau_i = \alpha' - \pi/3$,

$$\tau_i \ge \frac{\pi}{3} - 2a. \tag{3.5}$$

If $a \le \pi/6$, the cosine rule in the triangle $s_i p_{i+1} r_{i+1}$, using $|s_i - r_{i+1}| \ge 1$, gives

$$\cos h \le \frac{d^2 - 1 + 4\cos^2 a}{4d\cos a} \le \cos a.$$

For $d > 1$, the last inequality is equivalent to $d + 1 \le 4\cos^2 a$, and the case $d = 1$ is immediate. Hence $h \ge a$, and (3.4)–(3.5) yield $\tau_i + \tau_{i+1} \ge b$.

If $a \ge \pi/6$, it is enough, using $\tau_i \ge 0$, to prove $a + h \ge \pi/3$. The same cosine-rule bound reduces this to

$$d^2 - 1 + (2-d)(1 + \cos 2a) \le \sqrt{3} d \sin 2a.$$

The left side is at most $d^2 - 3d/2 + 2$, while the right side is at least $3d/2$; the required inequality is therefore $(d-1)(d-2) \le 0$. □

The two inequalities above exclude both separated and adjacent pairs of breaks.

Lemma 3.9. *Among the six relations $s_i = r_{i+1}$, $1 \le i \le 6$, exactly one fails. Its index h belongs to $\{1, 2, 3\}$. Thus t_i is defined for every $i \in \{1, \dots, 6\} \setminus \{h\}$, and*

$$\tau_h + 2\tau_{h+1} + \tau_{h+2} \ge \frac{\pi}{3}, \qquad \tau_{h+1} > 0. \tag{3.6}$$

Proof. There cannot be two breaks. If they occur at $i < j$ with $i + 2 \le j$, then nonnegativity of the turns and two applications of (3.2) give

$$\sum_{k=1}^{9} \tau_k \ge \frac{1}{2}(\tau_i + 2\tau_{i+1} + \tau_{i+2}) + \frac{1}{2}(\tau_j + 2\tau_{j+1} + \tau_{j+2}) \ge \frac{\pi}{3},$$

a contradiction. If $j = i + 1$, apply (3.3) and its reflected form. Their sum is at least

$$\angle q_{i+1} p_{i+1} r_{i+1} + \angle s_{i+1} p_{i+2} q_{i+1} = \frac{1}{2}\left(\frac{\pi}{3} + \angle s_{i+1} q_{i+1} r_{i+1}\right).$$

The identity follows by summing the angles around q_{i+1} in the two isosceles triangles and the equilateral triangle $p_{i+1} p_{i+2} q_{i+1}$. Since r_{i+1}, s_{i+1} are distinct unit neighbours of q_{i+1}, the last angle is at least $\pi/3$, again contradicting the total-turn bound.

Lemma 3.6 shows that at least one of the first three relations fails. Hence there is exactly one break, at an index $h \in \{1, 2, 3\}$. The first assertion in (3.6) is (3.2). If $\tau_{h+1} = 0$, its left side is at most the total turn of the strip, which is strictly less than $\pi/3$; therefore $\tau_{h+1} > 0$. □

4. Lattice forcing and the final contradiction

The remainder of the proof has two purposes. We first show that the middle of the strip must be flat. We then use the expansion inequality at equality to force a finite triangular-lattice patch. Every independent set of six vertices below is used in one of two ways: an upper bound of seventeen on its neighbourhood gives an immediate contradiction, whereas an upper bound of eighteen is tight and forces every residual neighbour position counted in that bound. The final independent set is read from the resulting patch.

We first record the neighbourhood information used in the finite case analysis. In every count below, the displayed labels form only an upper bound for the possible neighbours; coincidences between labels can only decrease the cardinality.

Lemma 4.1. *For $1 \le i \le 8$,*

$$N(p_i) = \{p_{i-1}, p_{i+1}, q_{i-1}, q_i\}. \tag{4.1}$$

Moreover,

$$N(p_0) = \{p_{-1}, p_1, q_0\}, \qquad N(p_9) \subseteq \{p_8, q_8\} \cup U_9, \quad |U_9| \le 2, \tag{4.2}$$

and, for $1 \le i \le 7$,

$$N(q_i) \subseteq \{p_i, p_{i+1}, q_{i-1}, q_{i+1}, r_i, s_i\}, \tag{4.3}$$

where absent unit edges are omitted. Finally,

$$\big|N(t_5) \setminus \{t_4, t_6, q_5, q_6\}\big| \le 2, \tag{4.4}$$

and, if $h \in \{1, 2\}$,

$$\big|N(t_4) \setminus \{t_3, t_5, q_4, q_5\}\big| \le 2. \tag{4.5}$$

Proof. The four vertices displayed in (4.1) are distinct unit neighbours of p_i. Indeed, Lemma 3.2 separates p_{i-1}, p_{i+1}, while $q_{i-1} = p_i + \omega^2 e_{i-1}$ and $q_i = p_i + \omega e_i$; any unwanted coincidence would require a counterclockwise turn of at least $\pi/3$, contrary to (3.1). Since $\deg p_i = 4$, equality follows. The assertions at p_0, p_9 follow from their degrees and the forced neighbours. Equation (4.3) is Lemma 3.3. The last two bounds are applications of Lemma 3.5 to (t_4, t_5, t_6) and, when $h \le 2$, to (t_3, t_4, t_5). □

Lemma 4.2. *Unless both t_4t_5 and t_5t_6 are edges of G, there is an independent set S satisfying $6|N(S)| \le 17|S|$.*

Proof. All boundary–auxiliary separations below follow from Lemma 3.2; the few remaining pairs are indicated explicitly.

Case 1: neither edge is present. If neither t_4t_5 nor t_5t_6 is an edge, take $S = \{p_0, p_2, p_4, p_6, p_8, t_5\}$. The five boundary vertices have at most the fifteen neighbours

$$\{p_{-1}, p_1, p_3, p_5, p_7, p_9, q_0, q_1, \dots, q_8\}.$$

By (4.4), and because both t-edges are absent, t_5 adds at most two vertices. Hence $|N(S)| \le 17$.

Case 2: only t_4t_5 is present. Suppose $t_4t_5 \in E(G)$ and $t_5t_6 \notin E(G)$. Lemma 3.4 gives $\tau_5 = \tau_6 = 0$ and $\tau_7 > 0$, so $q_6q_7 \notin E(G)$. If $h \in \{1, 2\}$, take $S = \{p_0, p_2, p_5, q_3, q_6, t_4\}$. Let $z = t_2$ when $h = 1$, and let $z = r_3$ when $h = 2$. By (4.5), at most two neighbours of t_4 are not already listed below; every other neighbour of S lies in

$$\{p_{-1}, p_1, p_3, p_4, p_6, p_7, q_0, q_1, q_2, q_4, q_5, z, t_3, t_5, t_6\},$$

which has at most fifteen vertices. The only auxiliary pairs not covered immediately by Lemma 3.2 are q_3q_6, q_3t_4, and q_6t_4; they are separated by parts (4), (5), and the final assertion of Lemma 3.4. Thus $|N(S)| \le 17$.

If instead $h = 3$, take $S = \{p_0, p_3, p_6, q_1, q_4, t_5\}$. Here $\tau_4 > 0$, so $q_3q_4 \notin E(G)$. Lemma 3.2 separates q_1, q_4 and q_4, t_5; moreover, projecting

$$t_5 - q_1 = e_1 + e_2 + e_3 + e_4 + \omega e_5 + \omega e_6 - \omega e_1$$

on e_1 shows that $|t_5 - q_1| > 1$. Since t_5t_6 is absent, (4.4) leaves at most two neighbours of t_5 outside the following set, which contains every other neighbour of S:

$$\{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_1, t_1, r_4, t_4\},$$

so again $|N(S)| \le 17$.

Case 3: only t_5t_6 is present. Suppose $t_4t_5 \notin E(G)$ and $t_5t_6 \in E(G)$. Then $\tau_6 = \tau_7 = 0$ and $\tau_5 > 0$. If $h \in \{1, 2\}$, take $S = \{p_0, p_2, p_4, p_6, t_4\}$. The four boundary vertices have at most twelve neighbours. Both t_3t_4 and t_4t_5 are absent, so (4.5) adds at most two, and $|N(S)| \le 14$.

If $h = 3$, take $S = \{p_0, p_2, p_7, q_3, q_4, q_5\}$. The break gives $q_3q_4 \notin E(G)$, while $\tau_5 > 0$ gives $q_4q_5 \notin E(G)$; Lemma 3.2(4) separates q_3, q_5. A direct union of the neighbourhoods in (4.1) and (4.3) is contained in

$$\{p_{-1}, p_1, p_3, p_4, p_5, p_6, p_8, q_0, q_1, q_2, q_6, q_7, r_3, s_3, r_4, t_4, t_5\},$$

so $|N(S)| \leq 17$.

The sets of size six have at most seventeen neighbours, and the set of size five has at most fourteen. Each violates Lemma 2.1. □

We may therefore assume that $t_4t_5, t_5t_6 \in E(G)$. By Lemma 3.4,

$$\tau_5 = \tau_6 = \tau_7 = 0. \tag{4.6}$$

We shall repeatedly use the elementary identity

$$|r + s\omega|^2 = r^2 + rs + s^2 \qquad (r, s \in \mathbb{Z}). \tag{4.7}$$

In particular, $|r + s\omega| = 1$ exactly for $r + s\omega \in \{\pm 1, \pm\omega, \pm\omega^2\}$.

Normalize $p_4 = 0$ and $e_4 = e_5 = e_6 = e_7 = 1$. Then

$$\begin{gathered} p_5 = 1, \quad p_6 = 2, \quad p_7 = 3, \quad p_8 = 4, \\ q_4 = \omega, \quad q_5 = 1 + \omega, \quad q_6 = 2 + \omega, \quad q_7 = 3 + \omega, \\ t_4 = 2\omega, \quad t_5 = 1 + 2\omega, \quad t_6 = 2 + 2\omega. \end{gathered} \tag{4.8}$$

The four known neighbours t_4, t_6, q_5, q_6 of t_5 leave only the closed angular sector from $\pi/3$ to $2\pi/3$ for any further neighbour. Its endpoint positions are

$$x = 3\omega, \qquad y = 1 + 3\omega. \tag{4.9}$$

Lemma 4.3. *Both positions x and y are occupied. In particular,*

$$N(t_5) = \{t_4, t_6, q_5, q_6, x, y\}.$$

Proof. Two unit neighbours in a closed sector of width $\pi/3$ must occupy its two endpoints. Thus, if one of x, y is absent, the sector contains at most one neighbour outside $\{t_4, t_6, q_5, q_6\}$.

If $h \in \{1, 2\}$, take $S = \{p_0, p_3, p_6, q_1, q_4, t_5\}$. Lemmas 3.1, 3.2, and 3.4 handle every pair except q_1, t_5; for that pair,

$$t_5 - q_1 = e_1 + e_2 + e_3 + e_4 + \omega e_5 + \omega e_6 - \omega e_1$$

has projection greater than one on e_1. Let $w = s_1$ when $h = 1$, and let $w = t_1$ when $h = 2$. Under the contrary assumption, t_5 has at most one additional neighbour. Apart from that vertex, all neighbours lie in

$$\{p_{-1}, p_1, p_2, p_4, p_5, p_7, q_0, q_2, q_3, q_5, q_6, r_1, w, t_3, t_4, t_6\},$$

a set of at most sixteen vertices. Hence $|N(S)| \leq 17$, contrary to Lemma 2.1.

If $h = 3$, take $S = \{p_6, p_9, q_4, q_7, t_5\}$. Lemmas 3.2 and 3.4 prove independence; in particular, q_3q_4 is absent and $|q_7 - t_5| \geq \sqrt{3}$. Apart from the two unlisted neighbours of p_9, the second outward neighbour s_7 of q_7, and the at most one additional neighbour of t_5, every neighbour lies in

$$\{p_4, p_5, p_7, p_8, q_5, q_6, q_8, r_4, t_4, t_6\}.$$

Thus $|N(S)| \leq 14$, again contradicting Lemma 2.1. Therefore both x and y are occupied. The degree bound now gives the stated exact neighbourhood. □

The next independent set has at most eighteen neighbours. Expansion therefore forces equality, which determines the two missing directions at t_6.

Lemma 4.4. *The two positions*

$$u = 3 + 2\omega, \qquad v = 2 + 3\omega \tag{4.10}$$

are occupied, and

$$N(t_6) = \{t_5, q_6, q_7, y, u, v\}.$$

Proof. The set $F = \{p_0, p_2, p_4, p_7, q_5, t_6\}$ is independent. Using (4.1) and (4.3), all neighbours of $F \setminus \{t_6\}$, together with the four known neighbours q_6, q_7, t_5, y of t_6, lie in

$$\{p_{-1}, p_1, p_3, p_5, p_6, p_8, q_0, q_1, q_2, q_3, q_4, q_6, q_7, t_4, t_5, y\}.$$

The degree bound leaves at most two further neighbours of t_6, so $|N(F)| \leq 18$. Corollary 2.2 forces equality. In particular, both remaining neighbour positions at t_6 occur. Relative to t_6, the four known directions are $2\pi/3, \pi, 4\pi/3, 5\pi/3$; the only two additional directions separated by at least $\pi/3$ are 0 and $\pi/3$, yielding u, v. $\square$

Lemma 4.5. *The vertex u satisfies*

$$\left|N(u) \setminus \{t_6, q_7, v, q_8\}\right| \le 2.$$

Proof. The known directions from u to v, t_6, q_7 are $2\pi/3, \pi, 4\pi/3$. Any other unit neighbour lies in the complementary closed arc from $5\pi/3$ to $\pi/3$. If three occur, their directions must be exactly $5\pi/3, 0, \pi/3$. The first position is $4+\omega$, a unit neighbour of $p_8 = 4$. By (4.1), it must be one of p_7, p_9, q_7, q_8. It is distinct from p_7, q_7, and equality with $p_9 = 4 + e_8$ would require $e_8 = \omega$, impossible because e_8 is obtained from $e_7 = 1$ by a clockwise turn of less than $\pi/3$. Hence this position is q_8, leaving at most two neighbours outside the displayed set. $\square$

Lemma 4.6. *If $h \in \{1, 2\}$, then G contains an independent set of nine vertices with at most twenty-five neighbours.*

Proof. Take

$$S = \{p_0, p_2, p_5, p_8, q_3, q_6, t_4, y, u\}.$$

Lemmas 3.1 and 3.2 handle every pair not involving y, u. Relative to $p_4 = 0$, the earlier boundary vertices satisfy $\operatorname{Re} p_2 < -1$ and $\operatorname{Re} p_0 < -2$, whereas $\operatorname{Re} y = 5/2$ and $\operatorname{Re} u = 4$. The remaining pairs involving y, u and one of p_5, p_8, q_6, t_4 are nonunit triangular-lattice differences by the lattice identity (4.7). Finally, $|q_3 - p_4| = 1$, while $|y|, |u| > 2$, so the reverse triangle inequality separates q_3 from y, u. Thus S is independent.

Since $h \le 2$, the bound (4.5) applies. The point $x = 3\omega$ is a neighbour of $t_4 = 2\omega$ and is distinct from t_5, q_4, q_5. It is also distinct from t_3, because the strip formula and $p_4 = 0$ give $t_3 = \omega + \omega^2 e_3$, while $t_3 = x$ would imply $|\omega^2 e_3| = 2$. Thus x is one of the at most two neighbours counted in (4.5), and at most one additional neighbour of t_4 remains. The vertex y has the four known neighbours x, t_5, t_6, v, and therefore at most two further neighbours; Lemma 4.5 permits at most two additional neighbours of u. Let $z = t_2$ when $h = 1$, and let $z = r_3$ when $h = 2$. All fixed neighbours lie in

$$\{p_{-1}, p_1, p_3, p_4, p_6, p_7, p_9, q_0, q_1, q_2, q_4, q_5, q_7, q_8, t_3, t_5, t_6, x, v, z\},$$

which has at most twenty vertices. The three residual contributions add at most $1 + 2 + 2$, so $|N(S)| \le 25$. Since $6 \cdot 25 \le 17 \cdot 9$, this contradicts Lemma 2.1. $\square$

It remains to treat $h = 3$. Then $\tau_4 > 0$, so $q_3q_4 \notin E(G)$. Two further applications of tight expansion complete the lattice patch.

Lemma 4.7. *If $h = 3$, the four positions*

$$a = -1 + 3\omega, \qquad b = -1 + 2\omega, \qquad c = 1 + 4\omega, \qquad d = 4\omega \tag{4.11}$$

are occupied. Moreover, $b = r_4$ and $u = s_7$.

Proof. The set $T = \{p_5, p_8, q_6, t_4, y, u\}$ is independent, because every difference between two of its vertices is a nonunit triangular-lattice vector. Its fixed neighbourhood is contained in

$$\{p_4, p_6, p_7, p_9, q_4, q_5, q_7, q_8, t_5, t_6, x, v\},$$

a set of twelve vertices. The vertex t_4 has the four known neighbours q_4, q_5, t_5, x, the vertex y has the four known neighbours x, t_5, t_6, v, and Lemma 4.5 leaves at most two additional neighbours of u. Therefore $|N(T)| \le 18$. Corollary 2.2 gives the reverse inequality. If either t_4 or y had fewer than two further neighbours, the upper bound would be at most seventeen. Thus both have two.

At t_4, the four known directions are $4\pi/3, 5\pi/3, 0, \pi/3$; the two remaining directions are $2\pi/3, \pi$, giving a, b. At y, the four known directions are $\pi, 4\pi/3, 5\pi/3, 0$; the remaining directions $\pi/3, 2\pi/3$ give c, d.

The point b is a unit neighbour of q_4. It is distinct from p_4, p_5, q_5, t_4, and it cannot equal q_3 because q_3q_4 is absent. Thus (4.3) gives $b = r_4$. Similarly, u is a unit neighbour of q_7, distinct from p_7, p_8, q_6, t_6; it is distinct from q_8, since $q_8 = u$ would imply $|\omega e_8| = |-1 + 2\omega| = \sqrt{3}$. Hence $u = s_7$. $\square$

A final independent set of six vertices attains the expansion bound and forces the remaining neighbours of a.

Lemma 4.8. *The vertex a has degree six, and the three positions*

$$e = -1 + 4\omega, \qquad f = -2 + 4\omega, \qquad g = -2 + 3\omega \tag{4.12}$$

are occupied. In particular,

$$N(a) = \{t_4, x, b, e, f, g\}.$$

Proof. The set $S_a = \{p_6, p_9, q_4, q_7, t_5, a\}$ is independent. Lemmas 3.2 and 3.4 handle the pairs not involving a; the pairs involving a are nonunit lattice differences, except for ap_9, which is separated by $\operatorname{Re} p_9 > 9/2 > \operatorname{Re} a + 1$.

The vertices other than p_9, a have all their neighbours in

$$\{p_4, p_5, p_7, p_8, q_5, q_6, q_8, t_4, t_6, x, y, u, b\},$$

a set of thirteen vertices. The endpoint p_9 has at most two further neighbours, and a has the three known neighbours t_4, x, b and hence at most three further neighbours. Thus $|N(S_a)| \leq 18$. Corollary 2.2 forces equality, so a has three additional neighbours. The known directions from a to b, t_4, x are $4\pi/3, 5\pi/3, 0$; three further directions in the complementary arc must be $\pi/3, 2\pi/3, \pi$, yielding e, f, g. ☐

The forced points in the flat case are shown in Figure 2.

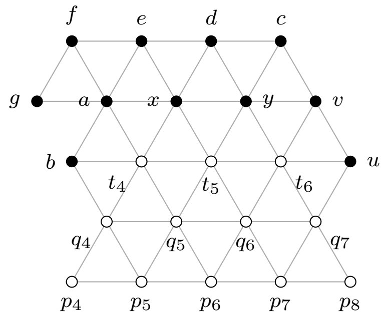

FIGURE 2. The triangular-lattice patch forced when $h = 3$. Open vertices belong to the original flat strip, and filled vertices are forced by the local expansion argument. For legibility, only selected unit adjacencies are drawn.

Lemma 4.9. *If $h = 3$, then G contains an independent set of eleven vertices with at most thirty-one neighbours.*

Proof. Take

$$S = \{p_0, p_2, p_6, p_9, q_3, q_4, q_7, t_5, a, v, d\}.$$

The relevant flat coordinates are displayed in Figure 2. The four boundary vertices are mutually nonadjacent, and Lemma 3.2 separates them from q_3, q_4, q_7, t_5. In the normalization (4.8), $p_2 = -(e_2 + e_3)$ and $p_0 = -(e_0 + e_1 + e_2 + e_3)$. Every one of these earlier edge vectors has real part greater than $1/2$, so $\operatorname{Re} p_2 < -1$ and $\operatorname{Re} p_0 < -2$. On the other hand, $\operatorname{Re} a = 1/2$, $\operatorname{Re} v = 7/2$, $\operatorname{Re} d = 2$, and $\operatorname{Re} p_9 > 9/2$. These projections separate p_0, p_2, p_9 from a, v, d, while the squared distances from $p_6 = 2$ to a, v, d are $9, 9, 12$.

The point q_3 lies on the unit circle about $p_4 = 0$. It is not adjacent to q_4, because $\tau_4 > 0$, while each of q_7, t_5, a, v, d has modulus greater than two; the reverse triangle inequality separates q_3 from those five vertices. Finally, the lattice identity (4.7) shows that the pairwise squared distances among q_4, q_7, t_5, a, v, d are all in $\{3, 9, 12\}$. Thus S is independent.

The vertex v has the four known neighbours t_6, y, u, c and at most two further neighbours. The point d has the four known neighbours x, y, c, e and at most two further neighbours. Using

(4.1)–(4.3), the exact neighbourhoods of q_4, q_7, t_5, a, and the two outward neighbours of q_3, all fixed neighbours of S lie in

$$\{p_{-1}, p_1, p_3, p_4, p_5, p_7, p_8, q_0, q_1, q_2, q_5, q_6, q_8, r_3, s_3,$$
$$b, t_4, t_6, x, y, u, c, e, f, g\},$$

which has at most twenty-five vertices. The two unlisted neighbours of p_9, the two possible extra neighbours of v, and the two possible extra neighbours of d give $|N(S)| \leq 25 + 2 + 2 + 2 = 31$. Since $6 \cdot 31 < 17 \cdot 11$, this contradicts Lemma 2.1. $\square$

Proof of Theorem 1.1. Let G be a smallest counterexample. Lemma 2.8 and Lemma 3.9 supply the nine-vertex strip and its unique break $h \in \{1, 2, 3\}$. Lemma 4.2 forces both t_4t_5 and t_5t_6 to be edges, so the strip is flat through p_8. Lemmas 4.3 and 4.4 force the first two additional lattice layers.

If $h \in \{1, 2\}$, Lemma 4.6 contradicts the deletion criterion. If $h = 3$, Lemmas 4.7 and 4.8 force the vertices used in Lemma 4.9, which gives the final contradiction. Thus no counterexample exists, and $\alpha(G) \geq 6|V(G)|/23$. Since the independence number is integral, the asserted ceiling follows. $\square$

Acknowledgements

The author acknowledges the use of OpenAI's ChatGPT. After the 7/27 proof had been completed, GPT-5.6-sol Pro was prompted specifically to improve that bound. It autonomously produced the central construction and proof strategy leading to 6/23, together with much of the exploratory work and substantial portions of the initial draft, and was subsequently used for simplification, cross-auditing, and expository revision. Earlier experiments with GPT-5.5 Pro were substantially more extensive and required considerably more time and manual effort, but did not improve the 7/27 bound. The author has repeatedly checked and revised the manuscript and takes full responsibility for the accuracy of its final contents.

References

[1] G. Csizmadia, *On the independence number of minimum distance graphs*, Discrete Comput. Geom. **20** (1998), 179–187.
[2] T. de Groot, *A 7/27 bound for penny graphs*, preprint, 2026, available online.
[3] P. Erdős, *Some combinatorial and metric problems in geometry*, in *Intuitive Geometry*, Colloq. Math. Soc. János Bolyai, vol. 48, North-Holland, Amsterdam, 1987, pp. 167–177.
[4] J. Pach and G. Tóth, *On the independence number of coin graphs*, Geombinatorics **6** (1996), 30–33.
[5] R. Pollack, *Increasing the minimum distance of a set of points*, J. Combin. Theory Ser. A **40** (1985), 450.
[6] K. J. Swanepoel, *Independence numbers of planar contact graphs*, Discrete Comput. Geom. **28** (2002), 649–670.